

INDIVIDUAL ASSIGNMENT 1

Course: Software Quality Assurance (SQA)

Unit Covered: Introduction to Software Quality Assurance

Instructions to the Student

- Answer all questions.
- Assignment must be handwritten

Section A: Short Answer Questions (20 Marks)

Question 1 (5 Marks)

Define the following terms:

- a) Software Quality
- b) Software Quality Assurance (SQA)
- c) Quality Control (QC)
- d) Reliability
- e) Maintainability

Question 2 (5 Marks)

State any five objectives of Software Quality Assurance.

Question 3 (5 Marks)

List and briefly explain five software quality attributes.

Question 4 (5 Marks)

Differentiate between:

- a) Software Quality Assurance and Software Testing
- b) Quality Assurance and Quality Control

Section B: Structured Questions (40 Marks)

Question 5 (10 Marks)

Explain the importance of Software Quality Assurance in modern software development. Support your answer with relevant examples.

Question 6 (10 Marks)

Discuss the evolution of Software Quality Assurance from the 1940s to the present day. Highlight the major milestones in its development.

Question 7 (10 Marks)

Describe the roles and responsibilities of the following stakeholders in Software Quality Assurance:

- a) Senior Management
- b) Project Manager
- c) Software Developer
- d) QA Engineer/Tester
- e) End User

Question 8 (10 Marks)

Software quality can be evaluated using both functional and non-functional quality dimensions.

Required:

- a) Explain Functional Quality.
- b) Explain Non-Functional Quality.
- c) Give three examples of each.

Section C: Applied Questions (40 Marks)

Question 9 (15 Marks)

A university has developed a new Online Student Registration System. Students have reported the following issues:

- Slow response during registration.
- Frequent system crashes.
- Difficulty navigating the interface.
- Occasional login failures.

Required:

- a) Identify the software quality attributes affected.
- b) Explain the impact of each issue on users.
- c) Suggest appropriate SQA activities that could prevent these problems.

Question 10 (15 Marks)

A software company wants to improve the quality of its products and development processes.

Required:

- a) Explain any three Software Quality Assurance Frameworks that can be adopted.
- b) Discuss the benefits of implementing these frameworks.
- c) Recommend the most suitable framework for a medium-sized software company and justify your answer.

Question 11 (10 Marks)

Case Study

An online banking application processes thousands of transactions daily. The organization wants to ensure that the system remains secure, reliable, and available at all times.

Required:

- a) Explain how Software Quality Assurance can help achieve these goals.
- b) Identify three quality attributes that are critical for this application.
- c) Recommend quality assurance activities that should be performed throughout the Software Development Life Cycle (SDLC).

END